

Fullstack Developer → *AI Engineer*

30

DAYS

An app-layer plan to **ship production LLM systems**—RAG, agents, evals—built on the skills you already have. No model training required.

Intermediate

2-3 hrs / day

Video · Projects · Reading

Four weeks. One *shippable* artifact each.

01 **Talk to the Models**
Ship one LLM feature in your own stack

02 **Retrieval (RAG)**
Chat-with-your-docs, with real citations

03 **Agents & Tools**
An agent that calls tools to finish a task

04 **Production**
Evaluated, deployed & monitored

You already know APIs, databases, async & deployment. This plan skips the relearning and goes straight to the AI layer—exactly what the role *is*.

WEEK 01

Talk to the Models

MILESTONE A working LLM feature wired into a real UI

Day 1–2 Prompt engineering for devs · structured output (Zod / Pydantic)

Day 3 Tokens, context windows, streaming & cost

Day 4 Build a streaming chat UI with the Vercel AI SDK

Day 5–7 Multi-provider (Claude · GPT · local) → ship the feature

Vercel AI SDK

Claude / GPT

Ollama

Zod

Pydantic

WEEK 02

Retrieval-Augmented Generation

MILESTONE “Chat with your docs” with source citations

Day 8–9 Embeddings, then build RAG *from scratch*—no framework

Day 10 Move to a real vector store (pgvector / Chroma)

Day 11–12 Chunking, reranking, hybrid search, citations

Day 13–14 Rebuild with LlamaIndex / LangChain → ship it

pgvector

ChromaDB

Embeddings

LlamaIndex

LangChain

Agents & Tool Use

MILESTONE A multi-step agent that calls tools & APIs

Day 15–16 Function calling · the ReAct reason→act→observe loop

Day 17 Stateful agent graphs with LangGraph

Day 18–19 Multi-tool, retries & fallbacks · connect via MCP

Day 20–21 Memory & light multi-agent → ship the agent

LangGraph

Function calling

MCP

CrewAI

Production

MILESTONE Evaluated, deployed & monitored—not a demo

Day 22–23 Build an eval set (LLM-as-judge) · add tracing

Day 24 Guardrails: validation & prompt-injection basics

Day 25 Cost & latency: caching, routing, streaming

Day 26–28 Deploy on Vercel / Railway · integrate the capstone

Evals

LangSmith

Guardrails

Caching

Write the story. Then *interview*.

Day 29 — Architecture diagram, README, demo GIF, push to GitHub. **Day 30** — Mock interview & plan the next 30 (fine-tuning, advanced RAG, multimodal).

IN ~70 HOURS YOU'LL HAVE

- A **deployed RAG + agent app** in your own stack
- A **GitHub portfolio piece** with a clear writeup
- Fluency across the full AI-engineering toolchain
- The system-design vocabulary to **interview**

THE TOOLKIT

Where to *learn* it.

► VIDEO

DeepLearning.AI short courses

Scrimba AI Engineer Path

Hugging Face LLM & Agents

☰ READING

Anthropic / OpenAI docs

LangGraph & LlamaIndex docs

DDIA (system design)

◇ TOOLS

Vercel AI SDK · pgvector

LangGraph · MCP · Chroma

LangSmith · Pydantic / Zod

Save it. Share it. Then go *build*.

The most competitive profile in 2026 isn't the person who studied AI—it's the engineer who shipped it.